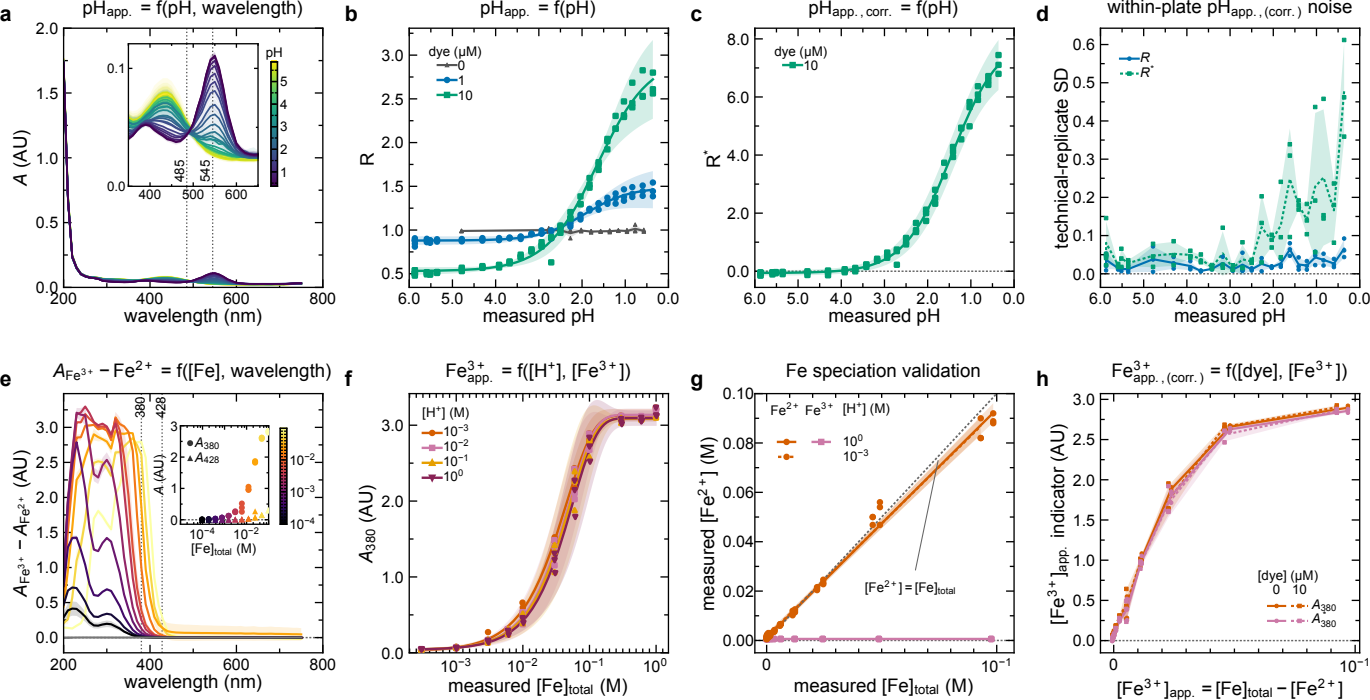

a
pH$_{app.}$ = f(pH, wavelength)
A (AU)
wavelength (nm)
pH
485
545
b
pH$_{app.}$ = f(pH)
dye (μM)
0
1
10
R
measured pH
c
pH$_{app.,corr.}$ = f(pH)
dye (μM)
10
R*
measured pH
d
within-plate pH$_{app.,(corr.)}$ noise
R
R*
technical-replicate SD
measured pH
e
$A_{Fe^{3+}} - Fe^{2+}$ = f([Fe], wavelength)
380
428
$A_{Fe^{3+}} - A_{Fe^{2+}}$ (AU)
wavelength (nm)
A_{380}
A_{428}
A (AU)
[Fe]$_{total}$ (M)
f
$Fe^{3+}_{app.}$ = f([H$^+$], [Fe^{3+}])
[H$^+$] (M)
10^{-3}
10^{-2}
10^{-1}
10^{0}
A_{380} (AU)
measured [Fe]$_{total}$ (M)
g
Fe speciation validation
Fe^{2+} Fe^{3+} [H$^+$] (M)
10^{0}
10^{-3}
measured [Fe^{2+}] (M)
[Fe^{2+}] = [Fe]$_{total}$
measured [Fe]$_{total}$ (M)
h
$Fe^{3+}_{app.,(corr.)}$ = f([dye], [Fe^{3+}])
[dye] (μM)
0 10
A_{380}
A_{380}
[Fe^{3+}]$_{app.}$ indicator (AU)
[Fe^{3+}]$_{app.}$ = [Fe]$_{total}$ − [Fe^{2+}]